



North Sydney Boys' High School
MATHEMATICS (HSC COURSE)
ASSESSMENT TASK 2 (2007)

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Name

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Mark

QUESTION 1

(a) Write down the primitive function of $\sqrt{x}$. **2**

(b) Evaluate:

i. $\int_0^1 (5x^4 - 3x^2 + 7) \, dx$ **2**

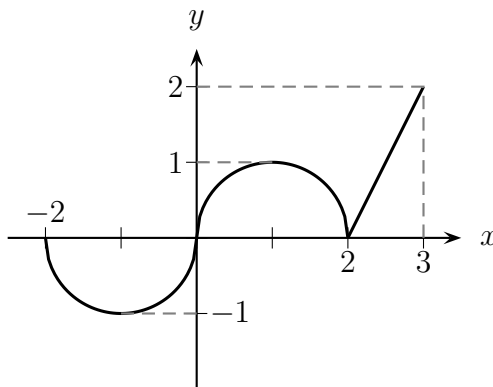
ii. $\int_{-1}^1 (2y - 1)^5 \, dy$ **3**

(c) The curve $y = f(x)$ has gradient function $\frac{dy}{dx} = 3 - 4x$. The curve passes through the point $(1, -1)$. Find the equation of the curve. **3**

(d) Find the value of A if **1**

$$2x^2 - 3x + 5 \equiv A(x - 1)^2 + x + 3$$

(e) The diagram illustrates a function $y = f(x)$ for $-2 \leq x \leq 3$. It consists of 1 line segment & 2 semi-circles. **2**



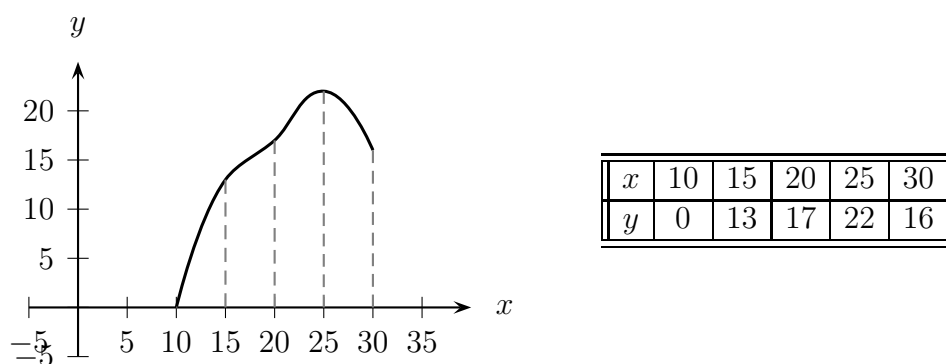
Evaluate $\int_{-2}^3 f(x) \, dx$.

QUESTION 2

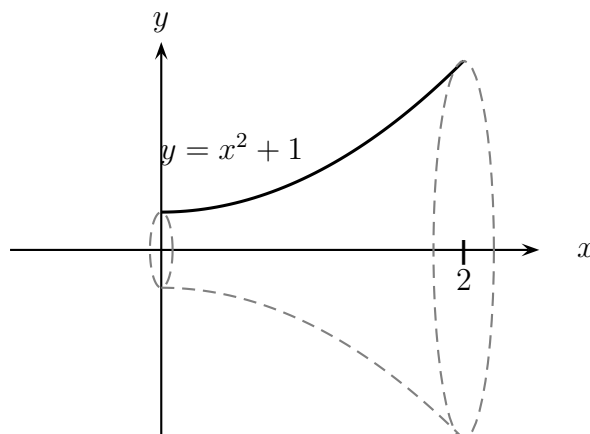
- (a) Sketch the locus of the point $P(x, y)$ which moves so that it is always a distance of 2 units from the point $(-2, 0)$. Hence write down its equation. 2
- (b) Solve the equation $x^4 - 7x^2 + 12 = 0$. 3
- (c) A point $Q(x, y)$ moves so that it is equidistant from the point $(1, 2)$ and the line $y = -2$. Describe the locus of point Q geometrically. (**Do not find its equation**). 1
- (d) If α & β are roots of the equation $2x^2 - 7x - 5 = 0$, find the values of:
- $\alpha + \beta$ 1
 - $\alpha\beta$ 1
 - $(\alpha + 1)(\beta + 1)$ 2
 - $(\alpha + 1)^{-1} + (\beta + 1)^{-1}$ 2

QUESTION 3

- (a) Find:
- $\int (2x - 1)(2x + 1) dx$ 2
 - $\int \left(\frac{2x^5 + 3}{x^5} \right) dx$ 2
- (b) Find the approximate area under the curve shown using the Trapezoidal Rule. 3



- (c) Find the volume of the solid of revolution shown below. 4

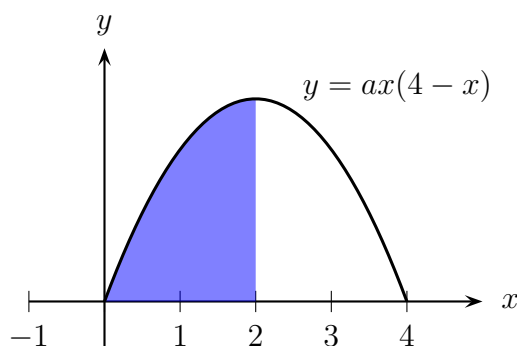


QUESTION 4

- (a) A parabola has vertex $V(3, 1)$ and directrix $y = -1$. 2
Find the equation of the parabola.
- (b) Find the values of k in the quadratic equation $x^2 - 5x + k - 1 = 0$ if:
- i. one root is equal to 2. 2
 - ii. one root is the reciprocal of the other. 2
- (c) For the parabola $x^2 = 8y - 24$, find the coordinates of the focus. 2
- i. Find the equation of the locus of point P . 3
 - ii. Describe this locus in geometrical terms, stating its important features. 2

QUESTION 5

- (a) Solve $(4 + k)(1 - k) < 0$. 1
- (b) For what values of k is the quadratic expression $kx^2 + 4x + (k + 3)$ positive definite? (Hint: part (a) may be useful). 3
- (c)
- i. Differentiate $(x^2 + 3)^5$. 1
 - ii. Hence, find $\int x(x^2 + 3)^4 dx$. 1
- (d) The area of the shaded region is 40 square units. 3



Find the value of a .

- (e) If $\int_{-1}^5 g(x) dx = 7$, find the value of:
- i. $\int_5^{-1} g(x) dx$. 1
 - ii. $\int_{-1}^5 [3g(x) + 2] dx$. 2

End of task

Brief solutions

QUESTION 1

(a) $\int x^{1/2} dx = \frac{2}{3}x^{3/2} + C$

(b) i. $\int_0^1 5x^4 - 3x^2 + 7 dx$
 $= [x^5 - x^3 + 7x]_0^1$
 $= 1 - 1 + 7 = 7$

ii. $\int_{-1}^1 (2y - 1)^5 dy$
 $= \left[\frac{(2y - 1)^6}{6 \times 2} \right]_{-1}^1$
 $= \frac{1}{12} ((2 - 1)^6 - (2(-1) - 1)^6)$
 $= \frac{1}{12} (1 - 729) = -\frac{728}{12}$
 $= -\frac{182}{3}$

(c) $y' = 3 - 4x$.

$$y = \int 3 - 4x dx = 3x - 2x^2 + C$$

At $x = 1$, $y = -1$.

$$\begin{aligned} -1 &= 3(1) - 2(1) + C \\ -1 &= 1 + C \\ \therefore C &= -2 \\ \therefore y &= 3x - 2x^2 - 2 \\ &= -2x^2 + 3x - 2 \end{aligned}$$

(d) $A = 2$:

$$\begin{aligned} 2x^2 - 3x + 5 &\equiv A(x - 1)^2 + x + 3 \\ &= Ax^2 - 2Ax - A + x + 3 \\ &= Ax^2 - x(2A - 1) + (3 + A) \end{aligned}$$

Equating coefficients, $A = 2$.

- (e) Note that the areas of the semi-circles are offset and the remaining area is the area of the triangle with length = 1 and height = 2.

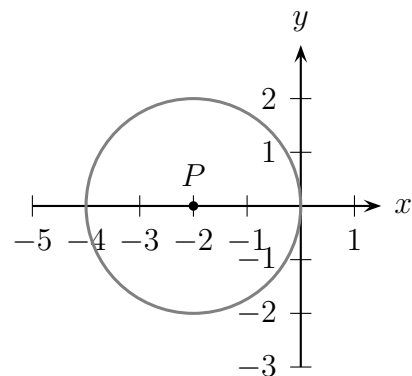
$$\int_{-2}^3 f(x) dx = \frac{1}{2} \times 2 \times 1 = 1$$

QUESTION 2

- (a) Using the distance formula to describe the distance from the point $P(x, y)$ to the point $(-2, 0)$ being 2 units,

$$2 = \sqrt{(x + 2)^2 + y^2}$$

$$4 = (x + 2)^2 + y^2$$



which is a circle centred at $(-2, 0)$ with radius 2.

- (b) Letting $m = x^2$,

$$m^2 - 7m + 12 = 0$$

$$(m - 4)(m - 3) = 0$$

$$(x^2 - 4)(x^2 - 3) = 0$$

$$(x - 2)(x + 2)(x - \sqrt{3})(x + \sqrt{3}) = 0$$

$$\therefore x = \pm 2, \pm \sqrt{3}$$

- (c) Parabola with focus at $(1, 2)$, vertex $(1, 0)$ and directrix $y = -2$.

(d) i. $\alpha + \beta = -\frac{b}{a} = \frac{7}{2}$

ii. $\alpha\beta = \frac{c}{a} = -\frac{5}{2}$

iii. $(\alpha + 1)(\beta + 1)$

$$= \alpha\beta + \alpha + \beta + 1$$

$$= \frac{7}{2} - \frac{5}{2} + 1$$

$$= 2$$

iv. $(\alpha + 1)^{-1} + (\beta + 1)^{-1}$

$$= \frac{1}{\alpha + 1} + \frac{1}{\beta + 1}$$

$$= \frac{\beta + 1 + \alpha + 1}{(\alpha + 1)(\beta + 1)}$$

$$= \frac{\alpha + \beta + 2}{(\alpha + 1)(\beta + 1)} = \frac{\frac{7}{2} + 1}{2}$$

$$= \frac{11}{4}$$

QUESTION 3

$$\begin{aligned} \text{(a)} \quad \text{i.} \quad & \int (2x-1)(2x+1) dx \\ &= \int 4x^2 - 1 dx \\ &= \frac{4}{3}x^3 - x + C \end{aligned}$$

$$\begin{aligned} \text{ii.} \quad & \int \left(\frac{2x^5+3}{x^5} \right) dx \\ &= \int \frac{\cancel{2x^5}}{\cancel{x^5}} + \frac{3}{x^5} dx \\ &= \int 2 + 3x^{-5} dx \\ &= 2x - \frac{3}{4}x^{-4} + C \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad A &\approx \frac{h}{2} (y_1 + \sum y_{\text{middle}} + y_\ell) \\ &= \frac{5}{2} (0 + 2(13 + 17 + 22) + 16) \\ &= 300 \text{ units}^2 \end{aligned}$$

$$\text{(c)} \quad \text{Since } y = x^2 + 1,$$

$$\begin{aligned} y^2 &= (x^2 + 1)^2 \\ &= x^4 + 2x^2 + 1 \\ V &= \pi \int_a^b y^2 dx \\ &= \pi \int_0^2 x^4 + 2x^2 + 1 dx \\ &= \pi \left[\frac{1}{5}x^5 + \frac{2}{3}x^3 + x \right]_0^2 \\ &= \pi \left(\frac{32}{5} + \frac{16}{3} + 2 \right) \\ &= \frac{206\pi}{15} \text{ units}^3 \end{aligned}$$

QUESTION 4

- (a) Focus-directrix form with (3, 1) being the vertex & $a = 2$ being the focal length as a is half of the distance between the focus & directrix.

$$\begin{aligned} (x-h)^2 &= 4a(y-k)^2 \\ \Rightarrow (x-3)^2 &= 8(y-1)^2 \end{aligned}$$

$$\text{(b)} \quad x^2 - 5x + (k-1) = 0.$$

- i. $\alpha = 2$. Using the sum of roots,

$$\begin{aligned} \alpha + \beta &= -\frac{b}{a} \\ 2 + \beta &= \frac{5}{1} \\ \therefore \beta &= 3 \end{aligned}$$

Using the product of roots,

$$\begin{aligned} \alpha\beta &= \frac{c}{a} \\ 2 \times 3 &= k-1 \\ \therefore k &= 7 \end{aligned}$$

- ii. $\alpha = \frac{1}{\beta}$. Using the product of roots,

$$\begin{aligned} \alpha\beta &= \frac{c}{a} \\ 1 &= k-1 \\ \therefore k &= 2 \end{aligned}$$

$$\text{(c)} \quad x^2 = 8y - 24$$

$$\Rightarrow x^2 = 4 \times 2(y-3)$$

i.e. vertex is at (0, 3) and focal length $a = +2$. Hence the focus is at (0, 5).

- (d) i. Applying the distance formula,

$$\begin{aligned} PA &= \sqrt{(x+3)^2 + (y-1)^2} \\ PB &= \sqrt{(x-3)^2 + (y+1)^2} \end{aligned}$$

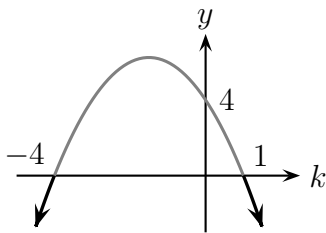
Applying the condition

$$\begin{aligned} PA^2 + PB^2 &= 70 \\ \overbrace{(x+3)^2 + (y-1)^2}^{PA^2} &+ \overbrace{(x-3)^2 + (y+1)^2}^{PB^2} = 70 \\ (x^2 + \cancel{6x} + 9) + (y^2 - \cancel{2y} + 1) &+ (x^2 - \cancel{6x} + 9) + (y^2 + \cancel{2y} + 1) = 70 \\ 2(x^2 + 9) + 2(y^2 + 1) &= 70 \\ x^2 + y^2 + 10 &= 35 \\ \Rightarrow x^2 + y^2 &= 25 \end{aligned}$$

- ii. Circle of radius 5 centred at the origin.

QUESTION 5

- (a) i. Sketching $y = (4 + k)(1 - k)$,



- ii. In $y = kx^2 + 4x + (k + 3)$,

$$\begin{aligned}\Delta &= b^2 - 4ac \\ &= 4^2 - 4k(k + 3) \\ &= 4(4 - k^2 + 3k)\end{aligned}$$

$y = kx^2 + 4x + (k + 3)$ is positive definite when $k > 0$ & $\Delta < 0$.

$$\begin{aligned}4(-k^2 + 3k + 4) &< 0 \\ (4 + k)(1 - k) &< 0\end{aligned}$$

Using Question 5(a)i,

$$k > 1 \quad k < -4$$

But it was established that $k > 0$. Hence $k > 1$ for the quadratic to be positive definite.

- (b) i. Differentiating by the chain rule,

$$\begin{aligned}\frac{d}{dx} (x^2 + 3)^5 &= 5 \times 2x (x^2 + 3)^4 \\ &= 10x (x^2 + 3)^4\end{aligned}\tag{5.1}$$

- ii. Using Equation (5.1),

$$\begin{aligned}\int x(x^2 + 3)^4 dx &= \frac{1}{10} \int 10x (x^2 + 3)^4 dx \\ &= \frac{1}{10} (x^2 + 3)^5 + C\end{aligned}$$

- (c) Integrating,

$$\begin{aligned}40 &= \int_0^2 4ax - ax^2 dx \\ &= \left[2ax^2 - \frac{a}{3}x^3 \right]_0^2 \\ &= (2a \cdot 4) - \left(\frac{a}{3} \cdot 8 \right) \\ &= \frac{16a}{3} \\ \therefore a &= \frac{120}{16} = \frac{15}{2}\end{aligned}$$

- (d) i. $\int_{-7}^{-1} g(x) dx = - \int_{-1}^5 g(x) dx =$
ii. $\int_{-1}^5 (3g(x) + 2) dx$

$$\begin{aligned}&= \int_{-1}^5 3g(x) dx + \int_{-1}^5 2 dx \\ &= 3 \int_{-1}^5 g(x) dx + \int_{-1}^5 2 dx \\ &= 3 \times 7 + 2 \times 6 \\ &= 33\end{aligned}$$